

Muhammad Humayun

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in Machine Learning, NLP, Computer Vision, Agentic AI, and LLM applications. Skilled in building end-to-end AI solutions using PyTorch, AWS, and Generative AI frameworks, with expertise in deep learning, CNNs, RAG systems, and AI agents. Passionate about developing scalable AI systems to solve real-world problems.

EDUCATION

University of Engineering & Technology Mardan

Bachelor of Science in Computer Science

Nov 2023 – Aug 2027

Mardan, PK

EXPERIENCE

InternPe

AI/ML Intern | Remote

Jan 2026 – April 2026

Arch Technologies

Generative AI Intern | Remote

Sep 2025 – Nov 2025

PROJECTS

• Multimodal Vision RAG

Multimodal RAG platform that combines GPT-4o Vision, FAISS semantic search, and FastAPI to enable intelligent querying over PDFs, images, and text documents through a ChatGPT-style React interface.

• LLaMA-7B QLoRA Fine-Tuning for Efficient LLM Adaptation

Fine-tuned LLaMA-7B using QLoRA on instruction-tuning datasets (Alpaca) for efficient parameter-efficient adaptation of large language models using PEFT and LoRA.

Reduced memory footprint using 4-bit NF4 quantization (bitsandbytes), enabling training on consumer-grade GPUs while freezing the base model and training only LoRA adapters.

• My-AI-Twin-Orchestrator (Production Deployed)

Developed production-grade AI Digital Twin with memory-enhanced conversations using AWS Bedrock, FastAPI, S3, and CloudFront.

Deployed end-to-end on AWS with RESTful API integration for scalable, context-aware AI interactions.

• Campus Marketplace — Flutter & Firebase Mobile Application

Built a full-featured campus buy-and-sell mobile application using Flutter and Firebase, enabling students to post, browse, search, and manage marketplace listings in real time.

Implemented Firebase Authentication, Firestore Database, Firebase Storage, image uploads, search functionality, user profiles, and contact integration through phone and WhatsApp.

Developed a scalable architecture using service-layer design, reusable widgets, and real-time Firestore streams for seamless user experience.

• AI-Powered Restaurant Automation System

Built fully automated restaurant support workflow using n8n, integrating OpenAI, Pinecone vector database, and ElevenLabs for AI-driven chat and voice interactions.

Implemented RAG-based context-aware responses for personalized customer service.

• AI-Powered Resume Screening Automation System

Built an end-to-end automation workflow using n8n, Gmail, Google Drive, and Google Sheets to collect and organize candidate resumes.

Implemented AI-based resume evaluation using OpenAI for scoring and hiring insights.

• AWS SageMaker End-to-End ML Pipeline

Built and deployed a complete ML pipeline on AWS SageMaker for mobile phone price classification including preprocessing, training, evaluation, and real-time inference deployment.

- **FraudShield — Real-Time Fraud Detection System**

Built an end-to-end fraud detection system using Random Forest with SMOTE for class imbalance. Deployed with FastAPI and Streamlit for real-time monitoring and prediction.

- **Stock Price Prediction with News Sentiment**

Built a hybrid system using FinBERT sentiment analysis and LSTM to predict next-day stock prices. Deployed with FastAPI and Streamlit with live news ingestion and dashboard.

- **Eye Disease Classification using EfficientNetB3**

Built a deep learning model using EfficientNetB3 for classifying retinal images into multiple eye diseases using transfer learning and data augmentation.

- **TrafficIQ — Real-Time Traffic Detection System**

Built a real-time traffic monitoring system using YOLOv8 for vehicle detection, tracking, and counting across multiple lanes. Developed FastAPI dashboard with WebSocket streaming and analytics.

- **Real-Time Face Recognition System**

Developed an end-to-end face recognition system using YOLOv8, FaceNet, and MTCNN for real-time identity detection. Built Flask dashboard with WebSocket streaming and SQLite database.

SKILLS

Programming Languages: Python, JavaScript, SQL, Dart

Data Science & Visualization : NumPy, Pandas, Matplotlib, Seaborn

ML / DL: PyTorch, TensorFlow, Scikit-learn, Deep Learning, CNNs, Transfer Learning, Natural Language Processing (NLP), Computer Vision

LLMs & Generative AI: LangChain, LlamaIndex, LangGraph, OpenAI API, LLaMA, Mistral, Ollama, Hugging Face Transformers, Prompt Engineering

RAG & AI Agents: RAG, Agentic RAG, Multi-Agent Systems, MCP, FAISS, Pinecone, ChromaDB, Weaviate, Embeddings, Semantic Search

Fine-Tuning & Optimization: LoRA, QLoRA, PEFT, bitsandbytes, Instruction Fine-Tuning, 4-bit NF4 Quantization

Automation & Workflows: n8n Automation, AI Workflow Automation, API Integrations

Cloud & MLOps: AWS (Bedrock, SageMaker, S3, CloudFront), Azure, Docker, CI/CD, GitHub Actions, LLMOps

Web Development: HTML, CSS, PHP

Mobile Development: Flutter, Firebase

APIs & Databases: FastAPI, Flask, MySQL, PostgreSQL, MongoDB, SQLite

Developer Tools: Git, GitHub, Google Colab, VS Code, Jupyter Notebook, Linux, Kaggle

Academic Foundations: OOP, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Software Engineering

Professional Skills: Problem Solving, Teamwork, Communication, Leadership, Critical Thinking, Agile Methodologies

CERTIFICATIONS

- Machine Learning Specialization – Stanford University / DeepLearning.AI
- MLOps Specialization – Udemy
- Generative AI Certification – Udemy